

## STATISTICAL ANALYSIS PLAN

Study Title: Survival Analysis of D-Penicillamine vs Placebo in Primary Biliary Cirrhosis (PBC Trial)

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### 1. STUDY OBJECTIVES

Primary objective:

To compare overall survival between patients treated with D-penicillamine and placebo in primary biliary cirrhosis.

Secondary objectives:

- To identify prognostic factors associated with mortality
- To assess the association between baseline bilirubin, albumin, age, and disease stage with survival outcomes

### 2. ANALYSIS POPULATIONS

Full Analysis Population (ITT):

All randomized patients (n=312) regardless of treatment compliance or discontinuation. Used for all efficacy analyses.

Safety Population:

All randomized patients who received at least one dose.

Exclusions:

106 non-randomized patients were excluded from all analyses.

### 3. PRIMARY ANALYSIS

Endpoint:

Overall survival, defined as time from randomization to death from any cause. Patients alive at last contact were censored.

Statistical method:

Survival will be estimated using the Kaplan-Meier method.

Differences between treatment groups will be assessed using the log-rank test. A two-sided p-value < 0.05 was considered to indicate statistical significance.

Multivariable analysis:

Cox proportional hazards regression adjusting for age, log-transformed bilirubin, albumin, and disease stage (as stratification variable). The proportional hazards assumption will be verified using Schoenfeld residuals (cox.zph).

#### 4. HANDLING OF MISSING DATA

Missing data mechanism:

Missing values in cholesterol and triglycerides are assumed to be Missing At Random (MAR), as missingness appears related to disease severity (bilirubin, stage) which are observed.

Method:

Multiple imputation by chained equations (MICE) with predictive mean matching (pmm). Number of imputations:  $m=5$ .

Predictors used: age, bilirubin, albumin, disease stage.

Results pooled using Rubin's rules.

Sensitivity analysis:

Complete case analysis will be performed as sensitivity analysis to assess robustness of results.

#### 5. SOFTWARE

All analyses were performed using R (version 4.4.x) with the following packages:

- survival: Kaplan-Meier and Cox regression
- survminer: visualization of survival curves
- tableone: baseline characteristics table
- mice: multiple imputation

All code is available at:

[github.com/anna-pharmwork/biostat-portfolio](https://github.com/anna-pharmwork/biostat-portfolio)

#### 6. RESULTS SUMMARY

Primary endpoint:

No statistically significant difference in survival was observed between treatment groups (log-rank  $p = 0.75$ ).

Multivariable Cox regression:

D-penicillamine was not associated with improved survival after adjustment for clinical factors (HR = 1.19, 95% CI [0.82, 1.71],  $p = 0.365$ ).

Prognostic factors:

- Log-bilirubin: HR = 2.60 per unit increase ( $p < 0.001$ )
- Albumin: HR = 0.40 per g/dL increase ( $p < 0.001$ )
- Age: HR = 1.03 per year ( $p < 0.001$ )
- Stage: used as stratification variable

Proportional hazards assumption:

Satisfied for all covariates after log-transformation of bilirubin and stratification by stage (global  $p = 0.81$ ).